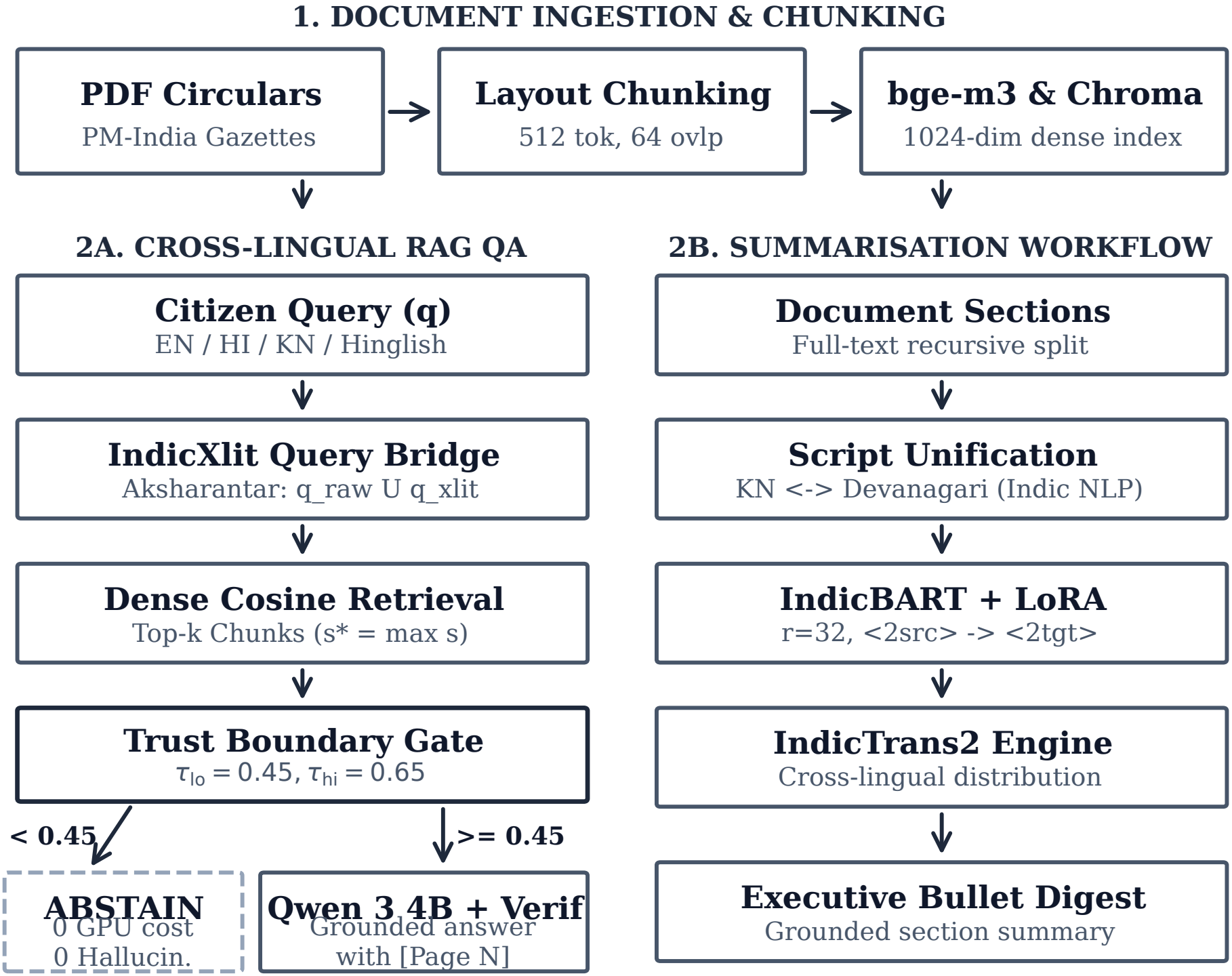
1. DOCUMENT INGESTION & CHUNKING
PDF Circulars
PM-India Gazettes
Layout Chunking
512 tok, 64 ovlp
bge-m3 & Chroma
1024-dim dense index
2A. CROSS-LINGUAL RAG QA
2B. SUMMARISATION WORKFLOW
Citizen Query (q)
EN / HI / KN / Hinglish
IndicXlit Query Bridge
Aksharantar: q_raw U q_xlit
Dense Cosine Retrieval
Top-k Chunks (s* = max s)
Trust Boundary Gate
$\tau_{lo} = 0.45, \tau_{hi} = 0.65$
< 0.45
>= 0.45
ABSTAIN
0 GPU cost
0 Hallucin.
Qwen 3 4B + Verif
Grounded answer
with [Page N]
Document Sections
Full-text recursive split
Script Unification
KN <-> Devanagari (Indic NLP)
IndicBART + LoRA
r=32, <2src> -> <2tgt>
IndicTrans2 Engine
Cross-lingual distribution
Executive Bullet Digest
Grounded section summary